

CARBOXYLATED NBR: PATENT LANDSCAPE AND SYNTHESIS/TERMINATION CONTROL IN EMULSION POLYMERIZATION

1. ABSTRACT

This report reviews the disclosed preparation of carboxylated nitrile rubber (carboxylated NBR; carboxyl group-containing nitrile rubber) from the provided patent set, with emphasis on direct synthesis and process control. The target material is defined as an acrylonitrile/butadiene-based rubber containing incorporated carboxyl functionality, typically introduced by copolymerization with a carboxyl-bearing monomer such as methacrylic acid. The single primary patent in the manifest, EP3604355B1, focuses not on the monomer feed itself but on how to terminate the polymerization and recover the latex without discoloration or post-storage instability. The disclosed route is a solvent-based polymerization of a monomer mixture comprising an α,β -ethylenically unsaturated nitrile monomer and a carboxyl group-containing monomer, followed by termination using a specific combination of nitrous acid salt and water-insoluble hydroquinone, then coagulation of the latex. The patent identifies that conventional terminators can lead to discoloration and poor storage stability, whereas the claimed terminator pair, used at controlled loading, suppresses conversion drift, Mooney viscosity increase, and discoloration. The examples compare multiple terminator combinations and show that nitrous acid salt plus water-insoluble hydroquinone gives low increases in conversion and viscosity after storage at 5°C and 50°C, with improved color. The broader technical trend is that carboxylated NBR performance is strongly influenced not only by monomer selection and incorporation of acid functionality, but also by late-stage polymerization termination chemistry and latex handling. For practical synthesis, the evidence supports using acrylonitrile with a carboxyl monomer such as methacrylic acid, terminating at a predetermined conversion with ≤ 0.15 parts by weight nitrous acid salt per 100 parts monomer mixture, and 0.1–1.0 part by weight water-insoluble hydroquinone, followed by coagulation to isolate the solid rubber.

2. METHODOLOGY

PRIMARY PATENT EVIDENCE

Patent 1

- Patent Number: EP3604355B1
- Patent Title: Method for producing carboxylated nitrile rubber
- Assignee: Not disclosed
- Jurisdiction: EP
- Publication Year: 2021
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: The patent discloses a method for producing carboxyl group-containing nitrile rubber by polymerizing a monomer mixture containing an α,β -ethylenically unsaturated nitrile monomer and a carboxyl group-containing monomer in a solvent, then terminating the polymerization with a nitrous acid salt and a water-insoluble hydroquinone, and finally coagulating the resulting latex. The core contribution is the termination strategy, which is designed to reduce discoloration and improve storage stability of the carboxylated NBR latex and solid rubber. - Nitrous acid salt loading: 0.15 parts by weight or less per 100 parts by weight of monomer mixture — Claim 1 - Water-insoluble hydroquinone loading: 0.1 to 1.0 part by weight per 100 parts by weight of monomer mixture — Claim 3 - Nitrous acid salt to water-insoluble hydroquinone ratio: 1:0.5 to 1:100 — Description - Acrylonitrile: 25 parts — Comparative Example 1 - 1,3-Butadiene: 68 parts — Comparative Example 1 - Methacrylic acid: 7 parts — Comparative Example 1 - Sodium nitrite: 0.025 parts — Comparative Example 1 - Sodium nitrite: 0.05 parts — Comparative Example 1 - Sodium nitrite: 0.15 parts — Comparative Example 1 - Sodium nitrite: 0.1 parts — Comparative Example 1 - Sodium nitrite: 0.2 parts — Comparative Example 1 - 2,5-Di-*t*-amylhydroquinone: 0.4 parts — Comparative Example 1 - 2,5-Di-*t*-amylhydroquinone: 0.2 parts — Comparative Example 1 - 2,5-Di-*t*-amylhydroquinone: 0.1 parts — Comparative Example 1 - Hydroxylamine sulfate: 0.2 parts — Comparative Example 1 - Diethylhydroxyamine: 0.2 parts — Comparative Example 1 - Methylhydroquinone: 0.2 parts — Comparative Example 1 - pH of latex of carboxyl group-containing nitrile rubber: 3.8 to 5.6 — Comparative Example 1 - Polymerization conversion ratio after addition of polymerization terminator: 65% — Comparative Example 1 - Polymerization conversion ratio after addition of polymerization terminator: 80% — Comparative Example 1 - Increase in polymerization conversion ratio after storage at 5°C: 0.0 to 16.7% — Comparative Example 1 - Increase in polymerization conversion ratio after storage at 50°C: 0.0 to 32.5% — Comparative Example 1 - Increase in Mooney viscosity after storage at 5°C: 0.1 to 17.0 — Comparative Example 1 - Increase in Mooney viscosity after storage at 50°C: 0.0 to 43 — Comparative Example 1 - Scorch time (t_5): 9.1 to 11 min — Comparative Example 1

Relevant Experimental Evidence - Comparative Example 1: Multiple terminator systems were compared; the nitrous acid salt plus water-insoluble hydroquinone combination gave the best suppression of conversion drift, viscosity increase, and discoloration. - Comparative Example 1: Alternative terminators such as hydroxylamine sulfate, diethylhydroxyamine, and methylhydroquinone showed poorer storage stability and/or discoloration performance. - Comparative Example 1: The monomer feed used for the carboxylated nitrile rubber was

acrylonitrile/1,3-butadiene/methacrylic acid at 25/68/7 parts.

Technical Relevance This patent is directly relevant because it discloses a practical route to carboxylated NBR and identifies a termination/coagulation strategy that preserves color and stability. It is especially useful for manufacturing control of acid-functional nitrile rubber latexes intended for adhesive, seal, gasket, and hose applications.

3. CROSS-PATENT COMPARISON & SYNTHESIS TRENDS

- No qualifying primary patents were identified under the configured target material relevance criteria. Cross-patent quantitative trends were therefore not generated.

4. CONCLUSION

The evidence supports carboxylated NBR preparation by copolymerizing acrylonitrile, butadiene, and a carboxyl monomer such as methacrylic acid, then carefully terminating the latex polymerization before coagulation. The key process insight from EP3604355B1 is that a nitrous acid salt plus water-insoluble hydroquinone, used at low loading, improves termination efficiency while minimizing discoloration and post-storage increases in conversion and Mooney viscosity. Based on the disclosed data, the most defensible synthesis guidance is to keep nitrous acid salt at 0.15 parts by weight or less per 100 parts monomer mixture, use water-insoluble hydroquinone at 0.1–1.0 part by weight, and coagulate only after the latex has been terminated to a pH of 7 or below.

5. REFERENCES

- EP3604355B1 | Method for producing carboxylated nitrile rubber | Not disclosed | EP | 2021 | <https://patents.google.com/patent/EP3604355B1/en>